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QUANTITATIVE RESEARCH

Black-Karasinski Parameter Estimation *BDO UK | 2026*

- **Aim:** Investigate the BK short-rate model in the context of credit default intensity, following observation of calibration difficulties arising from data scarcity and market illiquidity.
- **Findings:** Designed a parameter estimation methodology for credit volatility and mean reversion speed based exclusively on historical credit spreads, exploiting the OU/AR(1) equivalence.
- **Impact:** *Working paper under review; methodology adopted as BDO firm standard.*

Hybrid Framework for Discount for Lack of Marketability *BDO UK | 2025*

- **Aim:** Following observation of naive practitioner treatment of illiquidity, resolve theoretical limitations of canonical DLOM models (Chaffe, Ghaidarov, Longstaff).
- **Findings:** Formulated a Hybrid Put framework encompassing the full extent of downside risk and lost upside potential without assuming perfect market-timing ability; derived analytic approximation and validated numerical performance.
- **Impact:** *Working paper under review; framework validated as best-performing across synthetic CME outage windows.*

SABR Interpolation Error *BDO UK | 2025*

- **Aim:** Following observation of model error from parameter treatment at non-tenor dates, identify and validate a superior alternative.
- **Findings:** Validated total-variance interpolation and recalibration as more accurate, consistently outperforming linear and cubic SABR interpolation with RMSEs approximately 1.7 and 1.9 times higher respectively.
- **Impact:** *Working paper under review; adopted in production, improving accuracy of independent model valuations.*

Hidden Losses during the CME Trading Halt *Independent Research | 2025*

- **Aim:** Following the 28 November 2025 CME trading outage, develop an estimation framework to quantify hidden losses from the interruption.
- **Findings:** Treated outage as a temporary illiquidity shock; benchmarked DLOM models against synthetic outages; estimated hidden losses of approximately $10M across NASDAQ and S&P e-mini futures.
- **Impact:** *Working paper available; validated the Hybrid Put as best-performing across synthetic outage windows.*

QUANTITATIVE DEVELOPMENT

Automated Derivatives Valuation and Risk Engine *BDO UK | 2026*

- **Aim:** Develop and productionise a fully automated derivatives valuation and risk engine.
- **Findings:** Modular Python and C++ infrastructure standardising pricing across diverse instruments.
- **Impact:** *Drastically improved team-wide efficiency, project timelines and turnover, and pitch competitiveness.*

Black-Karasinski Automation *BDO UK | 2026*

- **Aim:** Leverage OU/AR(1) equivalence towards an automated BK parameter estimation and risk metrics system.
- **Findings:** Designed and implemented an automated system sourcing Refinitiv historical credit spreads, estimating parameters, and generating model-specific risk metrics.
- **Impact:** *Addressed existing practitioner limitations, expanding advisory suite of offerings.*

Automated Convertible Bond Risk Metrics *BDO UK | 2026*

- **Aim:** Develop an automated system for parameter-specific risk sensitivity metrics for previously unaddressed parameters.
- **Findings:** Launched system minimising model risk and improving result integrity for convertible bond advisory.
- **Impact:** *Expanded advisory capability for previously unaddressed parameter risks.*

Automated Reporting System *BDO UK | 2025*

- **Aim:** Develop a system to automatically generate technical reports, expanding the automation pipeline.
- **Findings:** Code-based pipeline producing structured technical reports from valuation engine outputs.
- **Impact:** *Significantly improved team-wide efficiency, freeing junior time for high-complexity projects.*

MACHINE LEARNING AND SYSTEMATIC TRADING

Generating Directional Signal from IV Metrics *CQF | 2025*

- **Aim:** Treat option Greeks as latent state variables of a stochastic Markov process; develop an ML system to predict asset price directionality from near-expiry implied volatility metrics.
- **Findings:** SVM classifier trained on one-day-to-expiry Greeks achieved accuracy of 0.99 and Macro F1 of 0.97, yielding out-of-sample returns 2.1 times greater than buy-and-hold.

- **Impact:** *Illustrated the potential to exploit information encoded in option Greeks, directly inspiring the current research agenda.*

Synthetic Derivatives Exchange Arbitrage *Independent Research | 2024*

- **Aim:** Develop an automated trading strategy exploiting mispricing of derivatives across crypto-exchanges.
- **Findings:** Strategy identified arbitrage opportunities across exchanges for individual instruments and between instruments and their synthesised equivalents.
- **Further Research:** *Dynamic hedging extension in progress.*

Optimised Windows Momentum Strategy *Independent Research | 2024*

- **Aim:** Design an automated trading system using realised volatility to predict the optimal window configuration for a momentum strategy.
- **Findings:** Linear regression relating realised volatility to tested window-length combinations achieved approximately 1.7 times greater returns (21%) than buy-and-hold in live trading.
- **Further Research:** *CNN-based realised volatility estimation across varying granularities under development.*

PROFESSIONAL EXPERIENCE

Associate, Quantitative Research | BDO UK *2024 – Present*

- Researched, developed, and implemented quantitative models for derivatives valuation across fixed income, credit, commodity, and exotic structures (CLNs, TARFs, path-dependent payoffs) using Numerix, QuantLib, Python, and C++.
- Identified unaccounted illiquidity risk; provided research report on significance, impact, and quantification methods, leveraging option-theoretic models towards a DLOM methodology, ultimately generating new business.
- Drove independent model research, demonstrating inaccuracies in client methodology, leading to authorship of internal technical research reports and larger client mandates.
- Performed in-depth technical review and challenge of third-party market risk models, ECL models, and hedging strategies across vanilla and exotic instruments.
- Delivered team-wide training on stochastic calculus, option theory, numerical methods, and their applications.

Analyst, Quantitative Research | BDO UK *2023 – 2024*

- Built and validated pricing models for vanilla and exotic derivatives; developed Bloomberg and Refinitiv data pipelines underpinning the firm's automated valuation infrastructure.
- Contributed to quantitative model research and challenger methodology development under senior supervision.

Research Assistant, Quantitative Research | Cobb Harbour Investments *2022*

- Investigated corporate bond ETF market microstructure; contributed to ML-based quantitative tool development.

Research Intern, Mechatronics | Monterrey Technological Institute *2019*

- Contributed to the engineering and development of novel paraplegic support products.

Strategy Intern | General Motors *2018*

- Contributed to the design and evaluation of quantitative market analytics.

Founding Member | Nuestro Futuro, A.C. *2020 – Present*

- Founded civil association towards environmental rights litigation targeting the Supreme Court of Justice in Mexico.

EDUCATION

Certificate in Quantitative Finance (CQF) — CQF Institute & Fitch Learning *2025 – Present*
Applied Stochastic Calculus 93% · Machine Learning 86% · Econometrics & Optimisation 84%

BEng (Hons) Systems Engineering — University of Warwick *2019 – 2022*
Systems Modelling & Computation 85% · Engineering Mathematics 87% · Signal Processing 77%
Dissertation: Visualising the Effects of Uneven Structures on Highly Rotational Vortex Cores

International Honours Program — Stanford University *2018*
Highest Distinction in Mathematics · Highest Distinction in English

International Baccalaureate — Edron Academy British International School *2017 – 2019*
Higher Level Mathematics 6/7 · Higher Level Physics 6/7 · Higher Level Economics 6/7

SKILLS

Programming:	Python (fluent) · C++ (intermediate) · C (intermediate) · MATLAB (advanced) · VBA
Platforms:	Numerix (fluent) · QuantLib (advanced) · Bloomberg (advanced) · Refinitiv · Capital IQ
Quantitative:	Derivatives pricing · Stochastic calculus · Monte Carlo · Statistical ML · Time series
Languages:	English (bilingual) · Spanish (bilingual) · French (intermediate)
Other:	GRE Quantitative 167/170 · CPD: Equity Options, IR & FX Derivatives (Clos Capital, 2024)